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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025**25MCS126 / 25CYS121 – ETHICAL HACKING AND NETWORK DEFENSE**

(For MCS, CYS)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

	Marks	BT	CO*	PO*
1. a) What is ethical hacking? Explain the role of ethical hackers in the cyber security.	5	2	1	1
b) Explain the types of offline password attacks.	5	2	2	1
c) Explain the countermeasures that must be implemented to prevent the cracking of passwords by a hacker.	10	2	2	1
2. a) Explain the types of penetration testing.	10	2	1	1
b) What is a threat model? Explain the key steps in threat modeling process.	5	2	1	1
c) Explain the ping sweep techniques used in scanning phase.	5	2	2	1
3. a) Explain the role of red team, blue team and purple team in the cyber security.	5	2	1	1
b) What is enumeration? Explain the SNMP enumeration technique.	10	2	2	1
c) Explain the countermeasures that can be taken against port scanning technique.	5	2	2	1

Module - II

4. a) Explain different types of attacks that can occur at each layer of OSI model.	10	1	3	1
b) Explain the DHCP starvation attack with a neat diagram.	5	2	4	1
c) Explain session hijacking.	5	2	4	1
5. a) Outline the Denial of Service (DoS) attack. Explain the countermeasures that can be carried out against the DoS attack.	10	2	4	1
b) Explain the DNS spoofing and its types.	5	2	3	1
c) Explain the working of Address Resolution Protocol (ARP) and ARP poisoning sniffing technique.	5	2	3	1
6. a) Explain the functionality of different layers in OSI reference model.	10	2	3	1
b) Explain the most common web application threats and their countermeasures.	10	2	4	1

Module - III

7. a) Explain the application-aware firewalls and proxy firewalls.	10	2	5	1
b) What is snort? What are its basic usages? Explain different modes and rules in snort.	10	2	5	1
8. a) Compare Stateless vs. Stateful Packet Filtering Firewalls.	10	3	5	1
b) What is NAT? Why is it required? Explain its working.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
